

A180 – Information: The Winding Quantum $\Delta_w = 1$

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.1 The Unit

[K] The elementary information unit of this theory is the *winding quantum*

$$\Delta w = 1.$$

It is the smallest difference between two winding classes on T^4 . Half a winding quantum does not exist – the fundamental group $\pi_1(T^4) \cong \mathbb{Z}^4$ is integer-valued (A030).

[B] Three properties follow immediately, and they are the reason why this choice is to be preferred over any cell-based definition:

- **integer-valued** – no arbitrary fineness;
- **coordinate-free** – independent of origin, orientation, and parametrisation;
- **energy-independent** – the unit is the same at every scale, because it is topological, not metric.

The last property is the strongest. An information unit that depended on the energy would not be a unit but a scale choice.

.2 Why No Unit Cells

[B] The obvious alternative – dividing space into cells and counting information – fails at the same objections listed in A030: origin, orientation, coordinate dependence, and an entropy that depends on an arbitrary partition.

For the winding quantum all these objections are empty, because a winding class has no location. It is the property of a closed path, not of a region. And the torus is boundaryless, $\partial T^4 = \emptyset$ – a surface-based definition is likewise excluded because no surface is distinguished.

.3 Magnitude and Phase

[K] The mass operator splits into two channels (A070, A120): magnitudes and phases of the spectral coefficients. For the concept of information the separation is essential.

[B] A magnitude spectrum fixes the phase only up to an all-pass factor. There is therefore information that is *in principle* not visible in the magnitude channel – the proof stands in A120, where it is shown that θ is captured by no symmetric, counting functional.

[STIPULATION] From this follows a booking rule for the whole series: a statement that uses only magnitudes can make no claim about phase information – neither its existence nor its absence.

.4 The Connection to Landauer

[B] The Landauer principle links the erasure of one bit to a minimum energy $k_B T \ln 2$. In the language of this theory this is a statement of the *translation layer*: it contains k_B and T , two quantities that are bridge and dress by A080.

The winding quantum lies beneath it. It is the topological unit; the Landauer energy is what it costs in a particular thermal dress.

[B] The derivation. On the 4-torus $\pi_1(T^4) \cong \mathbb{Z}^4$: the first homotopy group is $\mathbb{Z}^4$, each direction carries discrete winding numbers. Erasing a bit corresponds to the transition $\Delta w = 1$ – a change of winding number by one in one direction. This transition is irreversible (type-II loss, A090): the winding number is a topological invariant; its loss cannot be undone without expenditure of energy.

The energy expenditure follows from the thermodynamic dress of this topological transition. A system in thermal equilibrium at temperature T must for an irreversible bit transition dissipate at least entropy $\Delta S = k_B \ln 2$ (two states, one erased). The minimum energy is:

$$E_L = T \cdot \Delta S = k_B T \ln 2.$$

The winding quantum $\Delta w = 1$ is the geometric carrier of this step; the Landauer energy is its thermodynamic dress. Beneath: a topological transition. Above: a thermal energy amount.

[K] Limits. The derivation presupposes that the bit transition may be identified with the winding quantum – that is the physical claim that is not reduced further. The Landauer energy $k_B T \ln 2$ is a lower bound; actual erasure operations cost more. Whether exactly $\Delta w = 1$ is the only possibility to erase a bit is the second open question (see below).

.5 The Reversal Test

[B] The projection chain (A090) is lossless in one direction and not in the other. For the concept of information this means: the question “how much information is lost?” has an answer only for type I and type II; for type III the answer is zero.

In unrolling (type I) the loss can be exactly named – it is the winding number, specifically one integer per direction. This is the only place in the series where an information loss can be *exactly* quantified, and it is, because the lost quantity is integer-valued.

.6 Verification Script

- Integer-valuedness, coordinate-freedom, and scale-independence of Δw ; countercheck that a cell-based entropy depends on the partition and a spectral one does not: `python/A_Serie_Skripte/a180_windungsquant.py`

.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$S_L = k_B \ln 2$	Landauer energy; minimum cost per erased bit
Δw	Winding change; topological discreteness of information change
$I_{\max}$	Maximum information at given energy
$E_L = k_B T \ln 2$	Landauer bound at temperature T

.8 Open Questions

First, the derivation presupposes that bit transition and winding quantum may be identified. That is the core physical claim; it is not reduced further. The derivation is carried out in A180 (above), but the claim that exactly $\Delta w = 1$ carries a bit is a stipulation – not forced by the topology alone.

Second, it has not been shown that $\Delta w = 1$ is the *only* invariant unit. Shown is that it is invariant and that cell-based alternatives are not. That is less.

Third, an entropy formula is missing. This series contains no written-out entropy as a function of the spectrum – only the finding that such a function would automatically be invariant. That is a programme, not a result.

.9 Supersedes

This document subsumes: Dok. 243, 255, 257 (information), Dok. 290 (magnitude/phase), Dok. 301 (winding quantum), Dok. 302 (unit cells and partition invariance).

.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.